



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
27-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

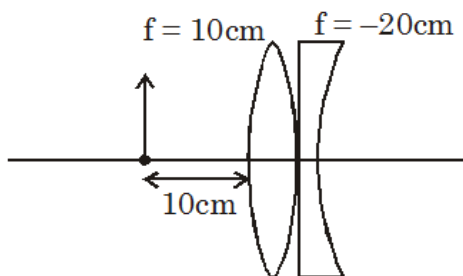
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Find the lateral magnification produced by the combination of lenses shown in the figure.



(A) 2

(B) 4

(C) 6

(D) 8

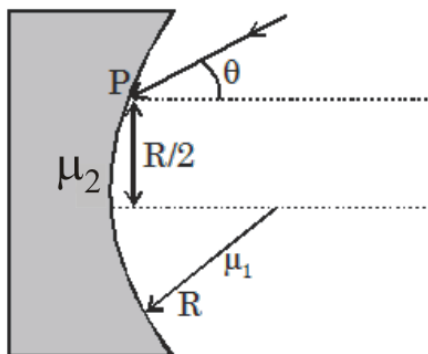
Space for Rough Work

2. A body is executing simple harmonic motion. At a displacement x , its potential energy is E_1 and at a displacement of y , its potential energy is E_2 . The potential energy E at a displacement $(x + y)$ is :-

- (A) $E_1 + E_2$ (B) $\sqrt{E_1^2 + E_2^2}$
 (C) $E_1 + E_2 + 2\sqrt{E_1 E_2}$ (D) $\sqrt{E_1 E_2}$

3. The minimum value of θ so that the ray gets totally reflected from spherical surface of radius of curvature

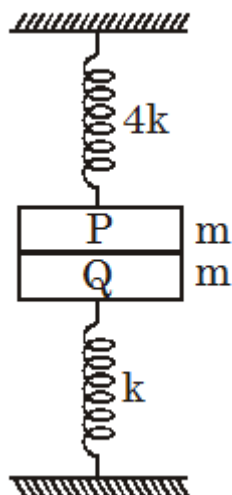
R is $\left[\frac{\mu_2}{\mu_1} = \frac{1}{2} \right]$



- (A) 30° (B) 60° (C) zero (D) not possible

Space for Rough Work

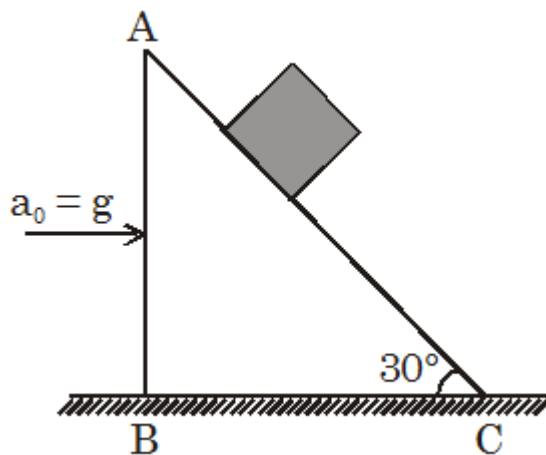
4. In the figure shown, blocks P and Q are in contact but do not stick to each other. The springs are in their natural lengths. The system is released from rest. Then the distance between P and Q when Q is at the lowest point for first time will be :-



- (A) $\frac{2mg}{K}$ (B) $\frac{4mg}{K}$ (C) $\frac{3mg}{K}$ (D) 0

Space for Rough Work

5. A block is placed on an inclined plane moving towards right horizontally with an acceleration $a_0 = g$. The length of the plane $AC = 1\text{ m}$. Friction is absent everywhere. The time taken by the block to reach from C to A is ($g = 10\text{ m/s}^2$) :-



(A) 1.2 s

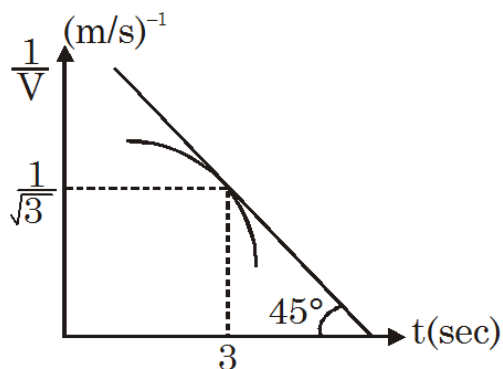
(B) 0.74 s

(C) 2.56 s

(D) 0.42 s

Space for Rough Work

6. The graph shows the variation of $\frac{1}{V}$ (where V is the velocity of the particle) with respect to time. Then find the value of acceleration at $t = 3$ sec.



(A) 5 m/s^2

(B) 1 m/s^2

(C) $3\sqrt{3} \text{ m/s}^2$

(D) 3 m/s^2

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

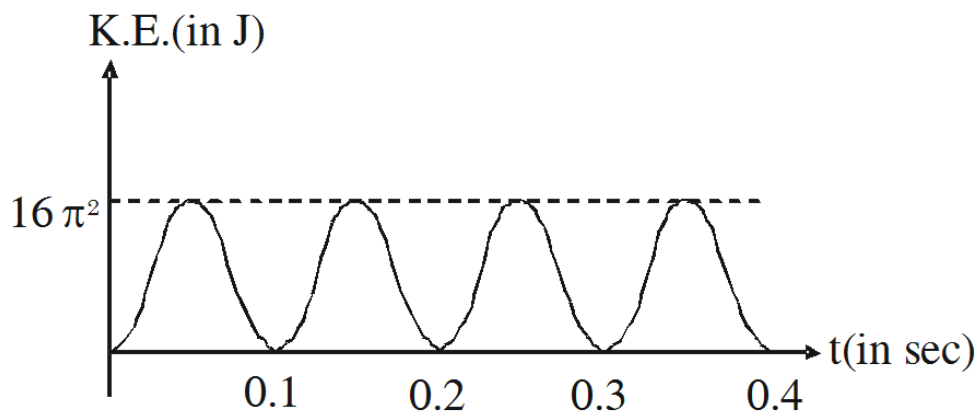
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
7. Consider the linear graph between magnification (m) and image position (v) for a lens. Which of the following statement about the straight line is correct ?
- (A) Focal length of the lens is equal to intercept on v -axis
- (B) Magnitude of focal length of the lens is equal to inverse of slope of the line.
- (C) Magnitude of intercept on m -axis is equal to unity
- (D) None of these
-

Space for Rough Work

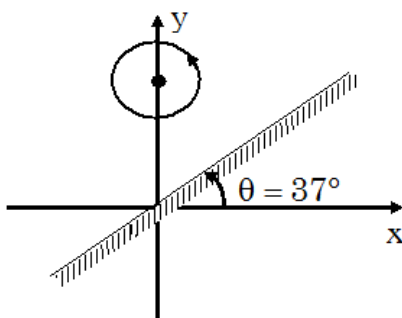
8. Kinetic energy of a particle executing SHM changes with time according to following graph. Mass of the particle is 2 kg. Then :-



- (A) amplitude of oscillation is 0.4 m
 (B) acceleration of the particle is zero at $t = 0.1$ s
 (C) time period of oscillation is 0.1 s
 (D) frequency of oscillation is 5 Hz.

Space for Rough Work

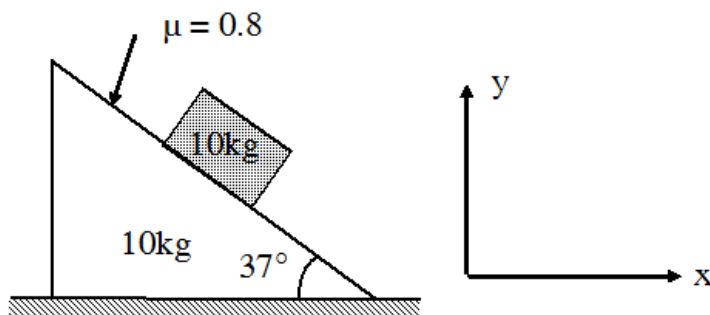
9. An object is moving in a circular path of radius 1m around (0, 10m) in xy plane with constant speed 1 m/s. Speed of image w.r.t. object is v .



- (A) $v_{\max} = 2$ m/s
 (B) $v_{\min} = 1$ m/s
 (C) v is minimum when object is closest or farthest from the mirror
 (D) v is maximum when object crosses y-axis
10. A particle of mass $m = 3\text{kg}$ is moving on a straight line in a conservative force field in which the potential energy associated with the particle is given by $U(y) = 2y^2 - 30y + 450$. At $t = 0$, particle is at $y = 8\text{m}$ and moving towards positive y-axis with 1 m/s.
- (A) During the course of its motion the maximum speed of the particle will be $\sqrt{\frac{4}{3}}$ m/s
 (B) Particle will never cross the origin.
 (C) Particle will come to its equilibrium position for the first time at $\frac{5\pi}{4\sqrt{3}}$ second.
 (D) The amplitude of oscillation for the particle is 1m.

Space for Rough Work

11. A wedge is moving rightwards on which a block of mass 10 kg is placed on it. Friction coefficient between the wedge and the block is 0.8. [take $g = 10 \text{ m/s}^2$]. Select correct alternative(s) among the following options.



- (A) If wedge is moving with constant velocity then acceleration of block is zero.
- (B) If wedge is moving with $\vec{a} = 2(\hat{i}) \text{ m/s}^2$ then friction acting on block is 44N.
- (C) If wedge is moving with constant velocity then friction acting on block is 64N.
- (D) If wedge is moving with $\vec{a} = 10(\hat{i}) \text{ m/s}^2$ then friction is 20N, downward on the block along the incline.
12. The instantaneous velocity v of a particle is related to its displacement x according to the relation $v = 8x + 1$ (All parameters in SI units). Which of the following statement(s) is (are) true if $x = 0$ at $t = 0$:-
- (A) The particle will experience a retardation.
- (B) The particle will be at rest at $t = 0$.
- (C) The displacement of the particle at time t is $x = \frac{1}{8}(e^{8t} - 1)$
- (D) The particle will experience an acceleration.

Space for Rough Work

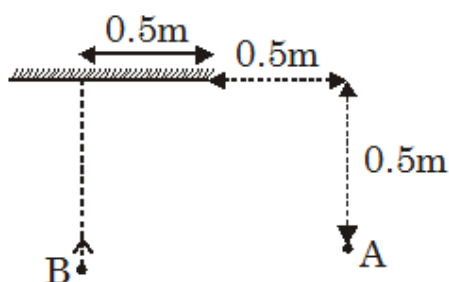
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

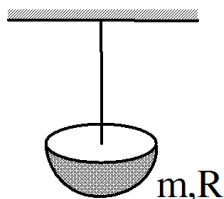
Zero Marks : 0 In all other cases.

1. A man 'A' stands at the position shown in the figure and a second man 'B' approaches the mirror along the line perpendicular to it which passes through its centre. At what distance (in meter) from the mirror will 'B' be at the moment when 'A' and 'B' first see each other in the mirror ?

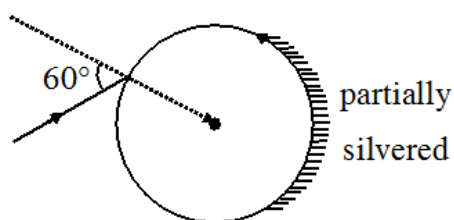


Space for Rough Work

2. A uniform solid hemisphere of mass m and radius R is attached to the roof with a chord of torsional constant C and performing torsional SHM. Then the time period (in seconds) of SHM is (Take $m = 15 \text{ kg}$, $R = \frac{2}{\pi} \text{ m}$, $C = 6 \text{ Nm/rad}$.)

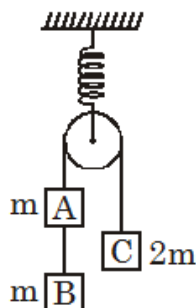


3. A ray is incident on a glass sphere as shown. The opposite surface of the sphere is partially silvered. If the net deviation of the ray transmitted at the partially silvered surface is $\frac{1}{3}$ rd of the net deviation suffered by the ray coming out after one reflection, the refractive index of the sphere is n . Find the value of n .



Space for Rough Work

4. Two particles are performing SHM with same amplitude and time period. At an instant two particles are having velocity 1 m/s but one is on the right and the other is on the left of their mean position. When the particles have same position, their speed is $\sqrt{3}$ m/s. Find the maximum speed (in m/s) of particles during SHM.
5. In the figure shown, pulley and spring are ideal and strings are light and inextensible. Initially all the bodies are at rest when string connecting A and B is cut. Find initial acceleration of C.



6. Two particles A and B located at points $(0, -10\sqrt{3})$ and $(0, 0)$ respectively in xy plane. They start moving simultaneously at time $t = 0$ with constant velocities $\vec{V}_A = 5\hat{i}$ m/s and $\vec{V}_B = -5\sqrt{3}\hat{j}$ m/s, respectively. Time when they are closest to each other is found to be K second. Find K. All distance are given in meter.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

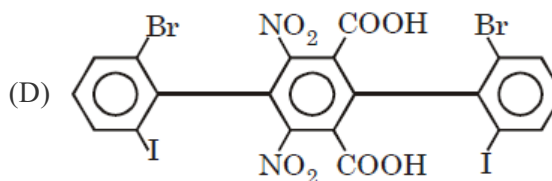
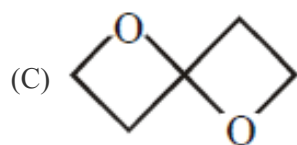
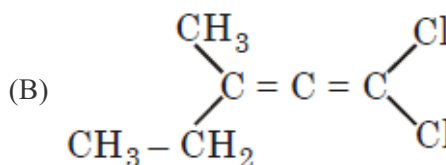
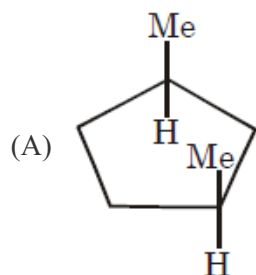
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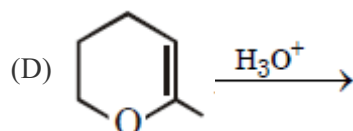
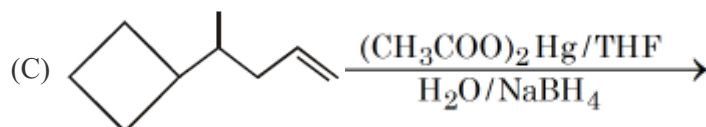
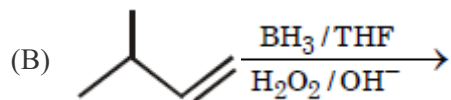
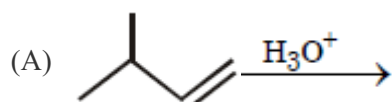
Negative Marks : -1 In all other cases

1. Which of the following molecule is resolvable / optically active?



Space for Rough Work

2. Which of following gives 2° alcohol major product?

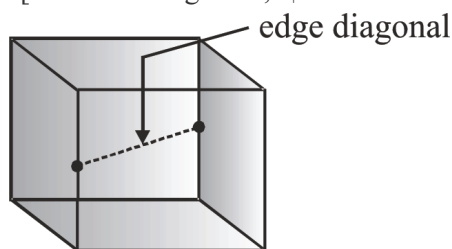


3. Assuming 100% ionization, the increasing order of the freezing point of the solution will be :-

- (A) $0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2 < 0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4 < 0.10 \text{ mol kg}^{-1} \text{ KCl}$
 (B) $0.10 \text{ mol kg}^{-1} \text{ KCl} < 0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4 < 0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2$
 (C) $0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4 < 0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2 < 0.10 \text{ mol kg}^{-1} \text{ KCl}$
 (D) $0.10 \text{ mol kg}^{-1} \text{ KCl} < 0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2 < 0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4$

Space for Rough Work

4. Correct expression for determining packing fraction (P.F.) of an ideal NaCl unit cell, if all ions along an edge diagonal are absent, as shown [Cl^- is forming FCC, r_+ = Cationic radius, r_- = Anionic radius]



- (A) $\frac{\frac{4}{3}\pi(r_+^3 + r_-^3)}{16\sqrt{2}r_-^3}$ (B) $\frac{\frac{4}{3}\pi\left(\frac{7}{2}r_+^3 + 4r_-^3\right)}{16\sqrt{2}r_-^3}$
- (C) $\frac{\frac{4}{3}\pi\left(\frac{5}{2}r_+^3 + 4r_-^3\right)}{16\sqrt{2}r_-^3}$ (D) $\frac{\frac{4}{3}\pi\left(\frac{5}{2}r_+^3 + 4r_-^3\right)}{16\sqrt{2}r_+^3}$

5. Which of the following statement is correct?

- (A) for $\text{N}_2 \rightarrow \text{N}_2^+$ bond order decreases and magnetic nature does not change.
- (B) for $\text{CO} \rightarrow \text{CO}^+$ bond order decreases and magnetic nature changes.
- (C) for $\text{O}_2 \rightarrow \text{O}_2^{2-}$ spin multiplicity decreases and magnetic nature changes.
- (D) for $\text{B}_2 \rightarrow \text{B}_2^+$ bond order increases and magnetic nature does not changes.

6. In which of the following maximum difference of % s-character between hybrid orbital of cationic and anionic part.

- (A) $\text{Cl}_2\text{O}_{6(s)}$ (B) $\text{N}_2\text{O}_{5(s)}$ (C) $\text{I}_2\text{Cl}_{6(l)}$ (D) $\text{PCl}_{5(s)}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

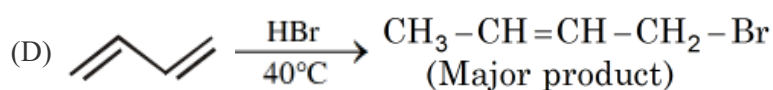
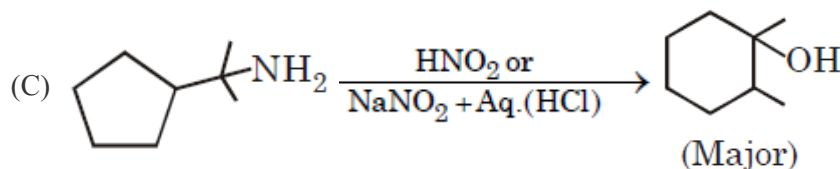
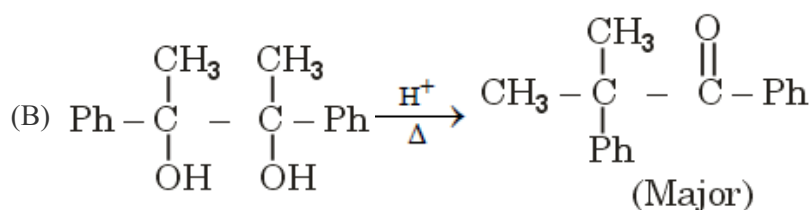
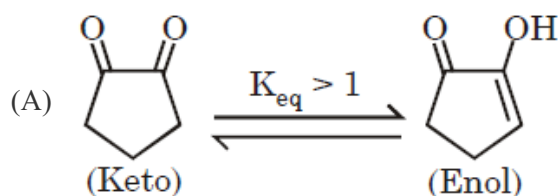
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

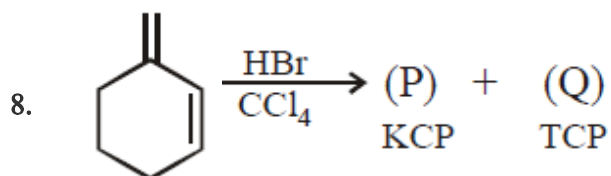
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

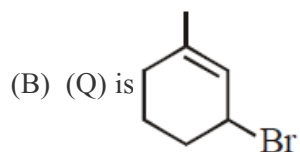
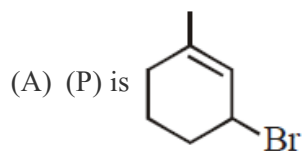
7. Which of following option is/are correct?



Space for Rough Work



Correct option(s) for above reaction sequence

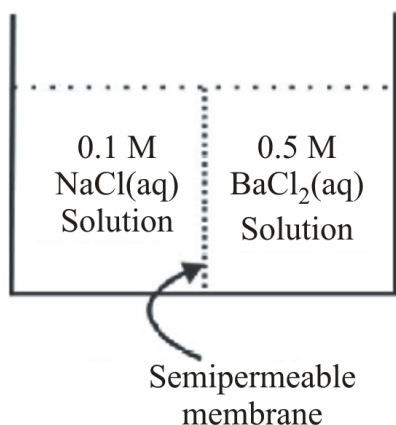


(C) On increasing the temperature % of (Q) increases in product mixture

(D) Product (P) is formed by more stable carbocation formation

Space for Rough Work

9. Study the following figure, and select the correct statements (assuming complete dissociation of the electrolyte).



- (A) There will be no movement of any solution across the membrane.
- (B) Water from BaCl₂ will flow towards the NaCl solution.
- (C) Water from NaCl will flow towards the BaCl₂.
- (D) The osmotic pressure of 0.1 M NaCl is higher than the osmotic pressure of 0.05 M BaCl₂.
10. Select the correct statement(s) :
- (A) The ionic crystal of AgBr may have Schottky defect
- (B) The unit cell having crystal parameters, $a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$ is hexagonal
- (C) In ionic compounds having Frenkel defect, the ratio r^+/r^- is high
- (D) The co-ordination number of Na⁺ ion in NaCl is 6

Space for Rough Work

11. Between ClO_2 and Cl_2O
- (A) The Cl - O bond length would be larger in Cl_2O
 - (B) The bond angle would be larger in Cl_2O
 - (C) ClO_2 is likely to be a coloured compound
 - (D) Cl_2O is likely to be soluble in water
12. Which of the following can be explained on the basis of Fajan's Rules.
- (A) Ag_2S is less soluble than Ag_2O .
 - (B) $\text{Fe}(\text{OH})_3$ is less soluble than $\text{Fe}(\text{OH})_2$.
 - (C) BaCO_3 is less soluble than MgCO_3 .
 - (D) BeCO_3 is thermally less stable than BaCO_3
-

Space for Rough Work

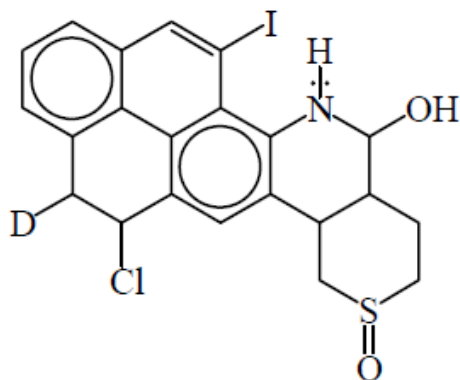
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. How many chiral center are present in this compound



2. An unknown unbranched hydrocarbon with molecular formula C_5H_{10} decolourise $Br_2 + H_2O$ and on reaction with $H-Cl/CCl_4$ give 2-chloro pentane as major product. Identify total number of possible reactants (excluding stereochemistry) -
3. A 1 litre solution containing equal moles of FeO and $Fe_{0.8}O$ was titrated with 70 ml 0.3M $KMnO_4$ in acidic medium. Millimoles of Fe^{3+} produced are -

Space for Rough Work

4. If shortest distance between two nearest Zn^{2+} ion in ZnS crystal is $3\sqrt{2}\text{\AA}$. Then edge length of this crystal (in $\text{\AA}$) will be
5. Find the no. of species having fractional bond order and paramagnetic in nature?
 N_2^+ , O_2 , O_2^+ , F_2 , B_2^+ , C_2^+ , CN^-
6. XY_n is an interhalogen compound which produces oxyacid of +5 oxidation state of central atom on complete hydrolysis. If 'P' moles H_2O is required for the complete hydrolysis of one mole XY_n then the value of $(n + P)$ is :

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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2$, then
 (A) $a = 1, b = 1$ (B) $a = 1, b = 2$ (C) $a = 1, b = -2$ (D) none of these
2. If $\tan(\cot^{-1}x) = \sin(\cos^{-1}x) + \cos(\sin^{-1}x)$, then value of $\sin^{-1}(x^2)$ is
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
3. If $f(x) = \begin{cases} -4 \sin x + \cos x & \text{for } x \leq -\frac{\pi}{2} \\ a \sin x + b & \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \cos x + 2 & \text{for } x \geq \frac{\pi}{2} \end{cases}$ is continuous, then
 (A) $a = -1, b = 3$ (B) $a = 1, b = 3$ (C) $a = 1, b = -3$ (D) $a = -1, b = -3$

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4. In the equation $2\log_e(2^x - 5) = \log_e 2 + \log_e \left(2^x - \frac{7}{2}\right)$ the value of x can be

(A) $\sqrt{3\sqrt{3\sqrt{3}\dots\infty}}$

(B) $\sqrt{6\sqrt{6\sqrt{6}\dots\infty}}$

(C) $\log_{5\sqrt{5}}(625)^{9/4}$

(D) $\left(2 + 2\sqrt{\log_2 3} - 3\sqrt{\log_3 2}\right)$

5. If $\lim_{x \rightarrow \infty} x^{1/3} \cdot (x^2 + 1)^{1/3} (ax^2 + bx + c)^{-1/2} = 2$, then the value of a is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

6. Evaluate $\lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{x}\right)^x x - ex \right)$

(A) e

(B) -e

(C) $\frac{e}{2}$

(D) $-\frac{e}{2}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Choose correct option(s)

(Where $[.]$ denotes greatest integer function)

(A) $\lim_{x \rightarrow 0} \left[\frac{\sin 2x}{x} \right] = 1$

(B) $\lim_{x \rightarrow 0} \left[-\frac{\tan^{-1} 3x}{x} \right] = -3$

(C) $\left[\lim_{x \rightarrow 0} \frac{\sin 3x}{x} \right] = 3$

(D) $\lim_{x \rightarrow 0} \left[\frac{\sin^{-1} x}{x} \right] = 0$

Space for Rough Work

8. Let $f(x) = (e^x + 2x)^{\frac{3}{x}}$, then

(A) $\lim_{x \rightarrow 0} f(x) = 9$

(B) $\lim_{x \rightarrow \infty} f(x) = e^3$

(C) $\lim_{x \rightarrow 0} f(x) = e^9$

(D) $\lim_{x \rightarrow \infty} f(x) = 3$

9. Which of the following pair of functions are **NOT** identical ?

(A) $f(x) = \sqrt{x^2}$ and $g(x) = (\sqrt{x})^2$

(B) $f(x) = \sqrt{\frac{x-2}{x-3}}$ and $g(x) = \frac{\sqrt{x-2}}{\sqrt{x-3}}$

(C) $f(x) = \tan x$ and $g(x) = \frac{1}{\cot x}$

(D) $f(x) = \frac{|x|}{x}$ and $g(x) = \operatorname{sgn}(x)$

10. If $f(x) = \sin\left(\ln\left(x + \sqrt{x^2 + 1}\right)\right)$ then $f(x)$ is -

(A) odd function

(B) symmetric about y axis

(C) domain is $\mathbb{R}$

(D) passes through origin

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11. Which of the following limits does not exist ? (where $[.]$ represents greatest integer function)

(A) $\lim_{x \rightarrow 1^+} ([x])^{\frac{1}{x-1}}$

(B) $\lim_{x \rightarrow 3} \frac{(x^2 - 9 - \sqrt{x^2 - 6x + 9})}{|x - 1| - 2}$

(C) $\lim_{x \rightarrow 0^+} (x)^{\ell_{\ln x}}$

(D) $\lim_{x \rightarrow 0^+} \left(\frac{1 - \cos(\sin^2 x)}{x^2} \right)^{\frac{\ell_{n(1-2x^2)}}{\sin^2 x}}$

12. Let $f(x) = \lim_{n \rightarrow \infty} [(1+x)(1+x^2)(1+x^4) \dots (1+x^{2^{2020n}})]$, $|x| < 1$, then

(A) $\sin^{-1} \sin f \left(\frac{1}{2} \right) = \pi - \frac{1}{2}$

(B) $\cos^{-1} \cos f \left(\frac{1}{3} \right) = \frac{3}{2}$

(C) $\tan^{-1} \tan f \left(\frac{1}{2} \right) = 2 - \pi$

(D) $\cos^{-1} \cos f \left(\frac{1}{2} \right) = 2$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Let N be the maximum value of $(\log_4 p_1 + \log_4 p_2 + \log_4 p_3 + \log_4 p_4)$ where $p_1, p_2, p_3, p_4 \in \mathbb{R}^+$. Also $p_1 + p_2 + p_3 + p_4 = 16$. Find the value of $\log_2 N^2$.
 2. The value of limit $\lim_{x \rightarrow 0} \frac{\sin(3x + a) - 3 \sin(2x + a) + 3 \sin(x + a) - \sin a}{x^3}$ is $-\cos\left(\frac{3a}{\lambda}\right)$. Then value of λ is (where $\lambda \in \mathbb{N}$)
 3. If the function $f : \mathbb{R} - \{-b\} \rightarrow \mathbb{R} - \{1\}$ is defined by $f(x) = \frac{x+a}{x+b}$ ($a \neq b$) is inverse of itself, then $\frac{b+2}{4}$ is
-

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4. If $f : [0, 3] \rightarrow [0, 5]$ is a bijective function defined by $f(x) = ax^2 + bx + 5$ (where $a, b \in \mathbb{R}$), then $f(3)$ equals
5. The least value of the expression $2\log_5 x - \log_x(0.04)$, for $x > 1$, is
6. Let $f(x) = \sin^{-1} \left(\frac{x+3}{2x+5} \right)$, $g(x) = \sin^{-1} \left(\frac{ax^2+b}{x^2+5} \right)$. If $\lim_{x \rightarrow \infty} (g(x) - f(x)) = \frac{\pi}{3}$ and $\lim_{x \rightarrow 0} (f(x) + g(x)) = \frac{\pi}{4}$ then value of $(a^2 + b^2)$
-

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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